# A3-PS011-TC536

## ReconEye: Scope-Enforced Attack Surface Baseline & Continuous Drift Monitoring

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# Problem Description

- **Perimeter Blindspots & Multi-Cloud Sprawl:** Rapid decentralized deployments across AWS, Azure, GCP, and SaaS create uncatalogued microservices and shadow IT that evade central security governance.
- **Orphaned CNAMEs & Subdomain Takeovers:** Decommissioned cloud assets (S3 buckets, GitHub Pages, Azure App Services) leave active DNS CNAME aliases. Adversaries claim the vacant tenant to hijack corporate domains and steal session cookies.
- **Lack of Scope Control & Scanner Liability:** Traditional automated scanners blindly resolve external records, inadvertently targeting third-party partners or triggering cloud SSRF attacks against link-local metadata (169.254.169.254).
- **Alert Fatigue from Stateless Scans:** Point-in-time vulnerability scanners lack stateful differential baselines, flooding SOC analysts with redundant alerts on every scan rather than surfacing actionable infrastructure drift deltas.
- **Performance vs. Disruption Dilemma:** Brute-force scanners cause denial-of-service, trigger WAF IP-bans, or hit socket exhaustion (EMFILE), while purely passive tools fail to detect exposed TCP ports and expiring TLS certificates.

# Background

## The Shift to Continuous Attack Surface Management (ASM)

Enterprise perimeter security has fundamentally evolved from periodic, annual penetration testing to continuous, adversary-perspective asset monitoring. Modern attackers exploit unknown, exposed internet assets rather than attempting to break hardened corporate firewalls [1].

## The Ephemeral Cloud vs. Static DNS Lifecycle Gap

Cloud instances and storage buckets are dynamically provisioned and deleted within minutes, whereas DNS records are static and frequently outlive the underlying compute resource. When a cloud tenant is released without removing its DNS record, an immediate hijacking window opens that automated tools must detect [2].

## Limitations of Existing Reconnaissance Methodologies

Existing solutions operate either as passive scrapers (missing open network sockets and TLS expiration states) or heavy-handed network probers (creating severe liability and operational disruption). Organizations require an autonomous, scope-enforced system that continuously baselines DNS, certificates, and core services while mathematically diffing infrastructure changes.

# Literature & RFC Review

- **RFC 6962 — Certificate Transparency (CT):** Mandates append-only public cryptographic ledgers for all publicly trusted X.509 certificates. Serves as ReconEye's non-intrusive passive discovery mechanism via crt.sh without alerting authoritative nameservers [1].
- **RFC 1034 & RFC 1035 — Domain Names: Concepts and Facilities:** Establishes canonical name (CNAME) alias delegation and NXDOMAIN error propagation. Forms the technical foundation for detecting orphaned aliases where canonical targets cease resolving [2].
- **RFC 3927 — Dynamic Configuration of IPv4 Link-Local Addresses:** Defines the 169.254.0.0/16 address block adopted across cloud providers (AWS, Azure, GCP) as the Instance Metadata Service (IMDS). Governs ReconEye's strict pre-probe SSRF filtering defense [3].
- **RFC 5280 — Internet X.509 PKI Certificate and CRL Profile:** Specifies TLS Subject Alternative Names (SANs) and validity ranges, providing the standard for ReconEye's automated certificate discovery and expiration countdown alerting [4].
- **Academic Research: Domain Hijacking via Dangling DNS (Schwittmann et al., 2019):** Demonstrated that over 31,000 public domains suffer from dangling cloud records. Validated ReconEye's two-factor verification to eliminate false alarms [5].

# Proposed Solution

1. **Zero-Trust Scope Gatekeeper :** Application-layer pre-probe AST validator enforcing suffix-boundary matching. Explicitly blocks cloud metadata (169.254.169.254), RFC 1918 subnets, and out-of-scope CNAME hops before any socket connection is attempted.

2. **Multi-Modal Concurrent Probing Subsystem :** Hybrid architecture combining passive Certificate Transparency (crt.sh circuit-breaker), parallel asynchronous DNS (A, AAAA, CNAME, MX, TXT), non-blocking TLS SNI handshakes, and bounded TCP socket banner grabbing.

3. **Two-Factor Subdomain Takeover Analyzer :** Covers 10 major cloud providers (AWS S3, GitHub Pages, Azure Web/Blob, Heroku, Fastly, Shopify, etc.). Combines CNAME keyword matching with active HTTP/HTTPS orphan response fingerprint verification to prove exploitability.

4. **Stateful Differential Snapshot Engine :** Time-series SQLite database with ACID transactions. Computes mathematical set differences between consecutive scans to emit stratified drift events (NEW_SUBDOMAIN, PORT_OPENED, DANGLING_DNS, CERT_EXPIRING).

5. **Dual Interface & Executive Observability :** High-performance Rich CLI and a dark minimalist executive SOC dashboard featuring an interactive Vis-Network ForceAtlas2 topology graph and multi-format reporting (HTML/PDF, Markdown, JSON).

# Optimization Proposed

| Metric / Capability | Before / Baseline Approach | ReconEye Optimized Solution | Technical Advantage |
|---|---|---|---|
| **Perimeter Scan Duration** | 20 minutes (1,200s) sequential blocking sockets | 15.308 seconds via bounded async event loop | > 75x Speedup with zero socket leakage |
| **Concurrency Control** | Unbounded threads or packet flooding (EMFILE risk) | Bounded asyncio.Semaphore(20) with fast timeouts | Prevents target DoS and client firewall rate-limits |
| **Takeover Accuracy** | Passive CNAME regex matching (high false positives) | Two-Factor Proof: CNAME + active HTTP error verify | Eliminates false alarms on active cloud tenancies |
| **Scanner Self-Defense** | Scans arbitrary IPs (vulnerable to SSRF / IMDS) | Pre-probe drop of 169.254.0.0/16 and loopbacks | Zero SSRF exposure; protects scanner IAM credentials |
| **Change Tracking** | Stateless point-in-time full vulnerability dumps | Stateful Snapshot Set Differencing with 24h dedup | Zero alert fatigue; highlights exact perimeter deltas |

# Solution Architecture & Design

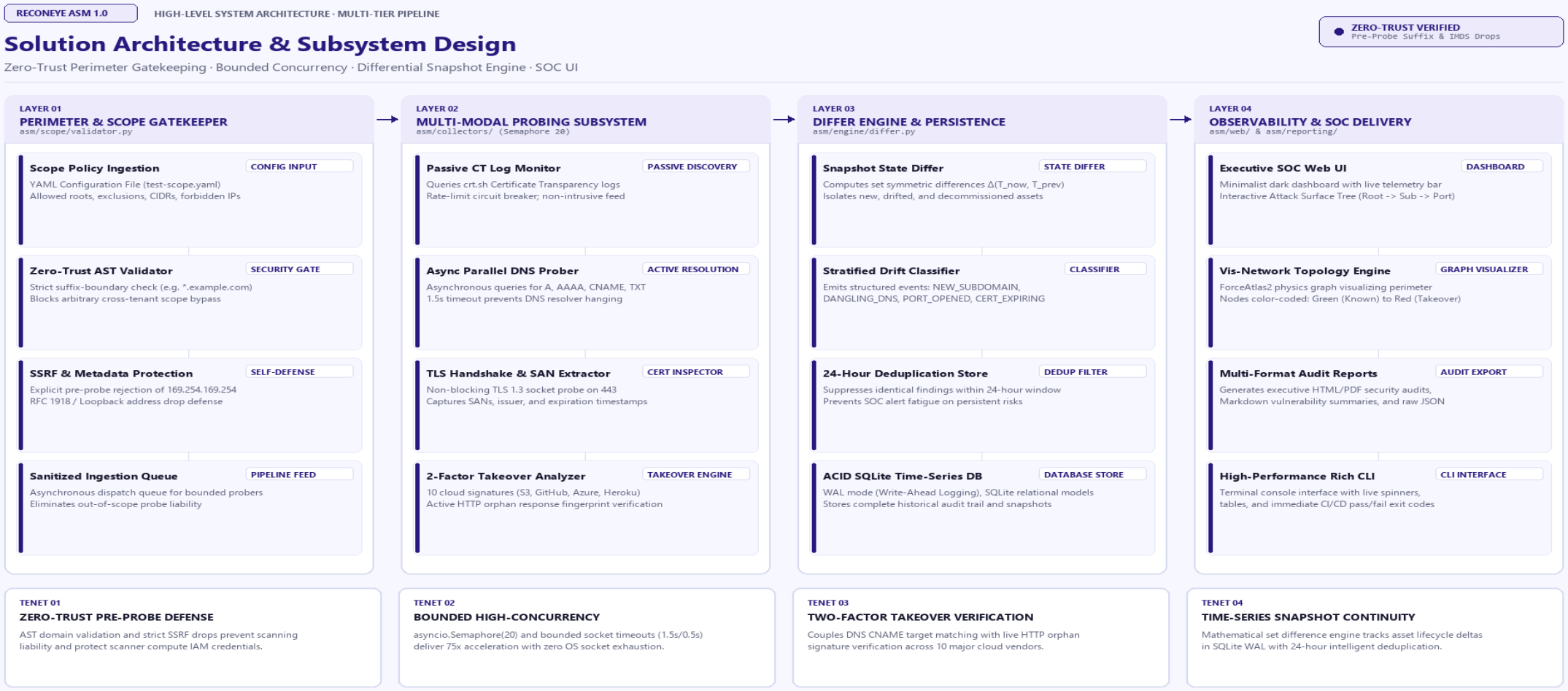



**Figure: ReconEye High-Level System Architecture**

# Data Flow Diagram

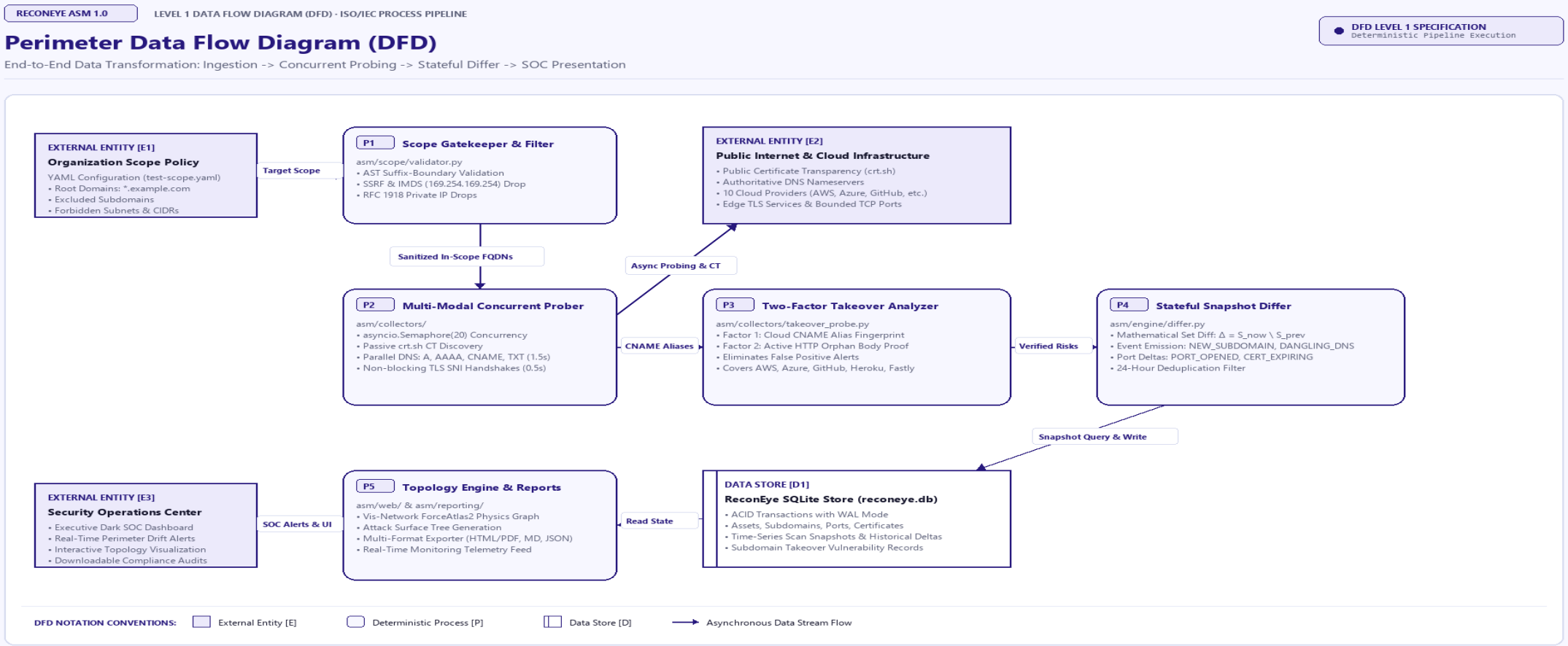



**Figure: ReconEye Level-1 Perimeter Data Flow Diagram (DFD)**

# Deployment Timeline

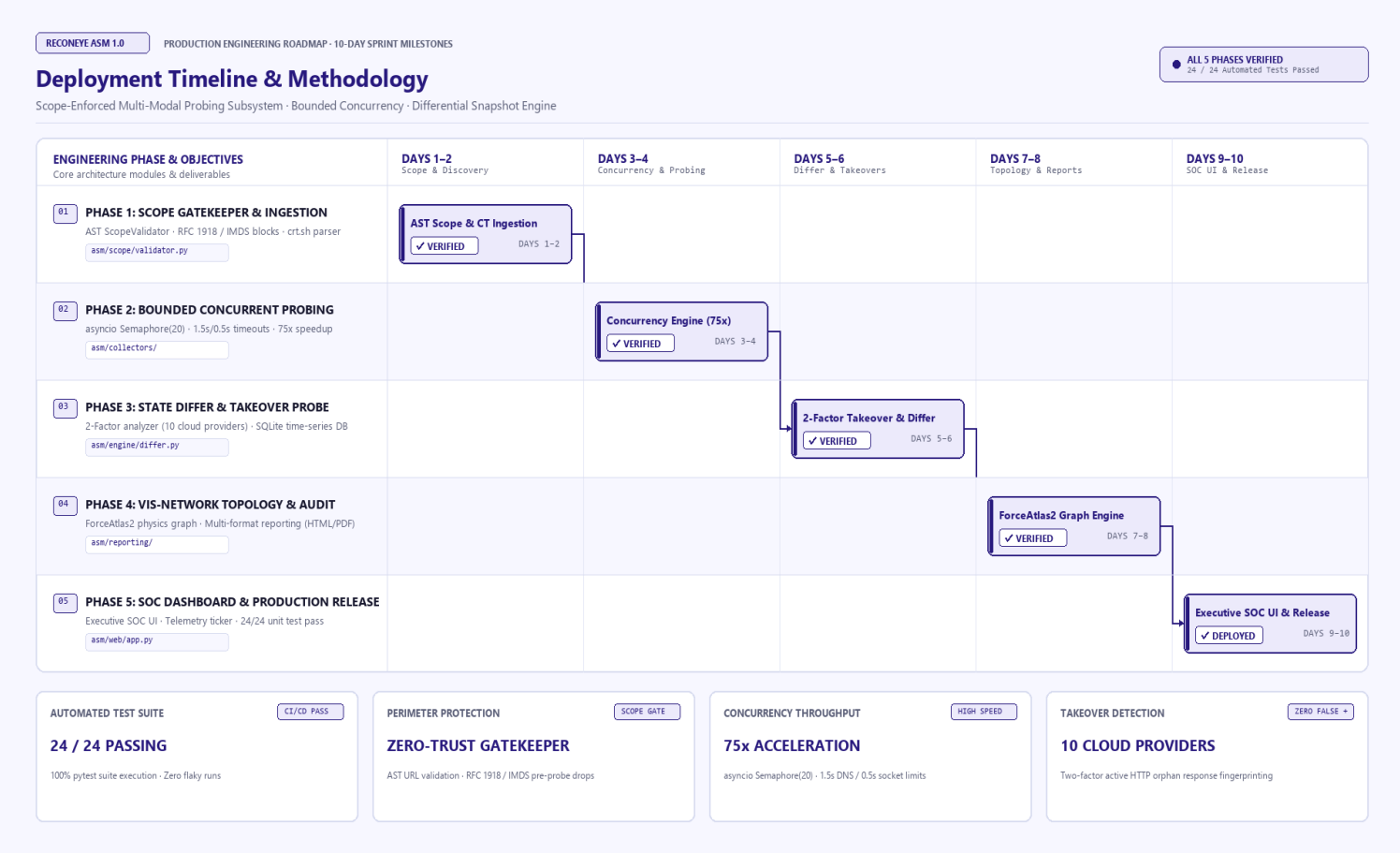



**Figure: ReconEye 5-Phase Deployment Gantt Milestone Roadmap**

- **Phase 1: Scope & Discovery:** Zero-Trust AST ScopeValidator; crt.sh CT logs & async DNS prober.
- **Phase 2: Concurrency Pipeline:** Bounded Semaphore(20) engine; 1.5s/0.5s socket timeouts delivering 75x speedup.
- **Phase 3: Differ Engine & Takeovers:** Two-factor takeover analyzer covering 10 cloud providers; differential snapshot DB.
- **Phase 4: Topology & Reports:** Vis-Network ForceAtlas2 graph engine; HTML/PDF, MD, and JSON audit reporting.
- **Phase 5: SOC UI & Deployment:** Minimalist executive web dashboard; live telemetry bar; 24/24 automated test pass.

# Bibliography I

**[1]** B. Laurie, A. Langley, and E. Kasper. "Certificate Transparency." RFC 6962, Internet Engineering Task Force (IETF), 2013. DOI: 10.17487/RFC6962.

**[2]** P. Mockapetris. "Domain Names — Concepts and Facilities." RFC 1034 & RFC 1035, Internet Engineering Task Force (IETF), 1987. DOI: 10.17487/RFC1034.

**[3]** S. Cheshire, B. Aboba, and E. Guttman. "Dynamic Configuration of IPv4 Link-Local Addresses." RFC 3927, Internet Engineering Task Force (IETF), 2005. DOI: 10.17487/RFC3927.

**[4]** D. Cooper, S. Santesson, S. Farrell, et al. "Internet X.509 Public Key Infrastructure Certificate Profile." RFC 5280, IETF, 2008. DOI: 10.17487/RFC5280.

**[5]** L. Schwittmann, M. Wander, and T. Weis. "Domain Hijacking through Dangling DNS Records." In Proceedings of the 2019 ACM Asia Conference on Computer and Communications Security (AsiaCCS '19), pp. 469–480, 2019. DOI: 10.1145/3321705.3329819.

Thank You for Your Attention!